3-D Joint Inversion of Airborne Electromagnetic and Magnetic Data Based on Local Pearson Correlation Constraints

Yunhe Liu, Xu Na, Changchun Yin, Yang Su, Siyuan Sun, Bo Zhang, Xiuyan Ren, and Vikas Chand Baranwal

***Abstract*— Based on the spatial structure correlation in different geophysical parameters, we propose a new 3-D joint inversion method for frequency-domain airborne electromagnetic (AEM) and airborne magnetic (AirMag) data by incorporating a local Pearson correlation constraint (LPCC). For each iteration, the entire model is separated into multiple subdomains and the Pearson correlation coefficients of resistivity and magnetization in the subdomain are employed as the additional regularization term to do the joint constraint. This new regularization term is continuously updated in the inversion process to ensure that the resistivity and magnetization models in two separated inversions converge to a similar spatial structure. As a statistics technology, the LPCC-based joint inversion scheme not only has the advantages of the conventional joint inversions, but also can implement the structural constraints in different scales by selecting different sizes of the subdomain. This provides the flexibility for solving multiscale problems. Synthetic examples show that the joint inversion can improve the overall inversion resolution by combining the high vertical resolution of the EM method and large exploration depth and high horizontal resolution of the magnetic method. In the application to field survey datasets, the joint inversion delivers better results than those of separate inversions, which further verifies the effectiveness of our method.**

***Index Terms*— 3-D joint inversion, airborne electromagnetic (AEM), airborne magnetic (AirMag), local Pearson correlation coefficient (PCC).**

I. Introduction

As EFFECTIVE exploration tools, airborne geophysics are widely used in the fields of geological mapping, hydrogeological survey, mineral exploration, and environmental monitoring [1]–[3]. The current airborne geophysical survey system can measure multiple data simultaneously, such as airborne electromagnetic (AEM) and airborne magnetic (AirMag) data, which improves the data acquisition efficiency and provides flexibility for joint inversion and interpretation [4]. The AirMag data contains deeper information and has a higher horizontal resolution than the AEM data. In addition, the magnetic data usually can directly indicate the mineral property.

Compared to AirMag data, the AEM data have better vertical resolution and can recover shallow geological structures more accurately. Thus, their joint inversion can improve the resolution [5]. However, at present, the inversion and interpretation of these two kinds of data are still done separately, which cannot integrate the resolution and detection ability of the two technologies. Motived by this situation, we propose to carry out 3-D joint inversion of AEM and AirMag data to pursue better data resolution to the underground structures of the earth.

The joint inversion means to invert the same underground geological structures using different types of geophysical data based on the correlation in petrophysical or geometric parameters. Vozoff and Jupp [6] first proposed to invert direct current (dc) resistivity method and magnetotelluric (MT) data. With the development of computational technology, the joint inversion has formed a series of algorithms and strategies, mainly including inverting the same physical parameters using different data, adding additional constraints to the similarity of spatial structures, and constructing function relationships between different physical properties, and so on. Inverting the same physical property is relatively simple, which is implemented by using data from different geophysical techniques. Yutaka and Sasaki [7] realized the joint inversion of marine controlled-source EM (CSEM) and MT-sounding data. Meqbel and Ritter [8] tested the joint inversions with CSEM, MT, and dc resistivity data. Using constraints of spatial similarity for different physical parameters is the mainstream technology of joint inversion at present. Haber and Oldenburg [9] proposed the structural coupling constraint for the joint inversion, in which they implemented the consistency constraint by minimizing the curvature difference between different models. The cross-gradient constraints proposed by Meju [10] and Gallardo and Luis [11] assume that the target body has obvious

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and consistent boundaries in different geophysical fields. The structural changes of different models are constrained as similar as possible by minimizing the vector product of model gradients. This method has been widely used in geophysical data inversion [12]–[14]. Zhdanov *et al.* [15] proposed a joint inversion method based on Gramian constraints, which are also structural similarity constraints. This method measures the correlation in the distribution of different physical parameters by transforming the model into the Gramian space. Moreover, Lelièvre *et al.* [16] leveraged the fuzzy c-means (FCM) clustering method to constrain the distribution of physical properties in the parameter domain and to make it distributed around the clustering center. Sun and Li [17], [18] used the improved FCM clustering method to realize the multidomain guided aggregate joint inversion for gravity and seismic data. Unlike the above-mentioned technologies, Jegen *et al.* [19] and Heincke *et al.* [20] introduced the petrophysical relationship between different physical properties in the joint inversion. In the inversion process, one transforms one physical property into another one, so one needs only to invert one kind of parameter. This kind of method requires a large number of petrophysical experiments as the basis to construct the function relationship between different parameters. This is difficult to carry out in most cases since the relationship between the two parameters is difficult to determine.

Compared with the ground geophysical explorations, although airborne geophysics has a high degree of technical integration and mature routine for data interpretation, there are few studies on the joint inversion for multiple parameters. To achieve the joint inversion of AEM and AirMag data using the structural similarity between different physical parameters, we propose a new scheme based on the local Pearson correlation constraints (LPCCs). The Pearson correlation coefficient (PCC) is usually used to measure the linear correlation between two variables in statistics. One of its important mathematical characteristics is that the change of the scales and positions in two types of data will not lead to the change of the PCCs. Thus, the PCC is very suitable for evaluating the linear correlation between parameters in different magnitudes and dimensions [21]. Different from the conventional joint inversions, the LPCC can be implemented in different scales, making it be able to solve multiscale problems.

In this article, we first introduce the joint inversion algorithm that uses the alternating iterative technique [22]. Then, we address in detail the calculation of PCC and the gradient of the objective function, and the inversion strategy adopted in this article. Finally, we demonstrate the effectiveness and practicability of our 3-D joint inversion based on LPCC through numerical experiments on both synthetic and field survey data from Lofoten–Vesteralen, Northern Norway.

II. Methodology

A. Objective Function for Joint Inversion

According to Kelbert and Egbert [23] and the joint inversion strategy proposed by Gao and Zhang [22], we define the objective function for our 3-D joint inversion as

$$\varphi_{\mathrm{EM}}(\boldsymbol{m}_1,\boldsymbol{m}_2)=\varphi_{d_1}(\boldsymbol{m}_1)+\lambda_1\varphi_{\boldsymbol{m}_1}(\boldsymbol{m}_1)+\gamma_1\varphi_J(\boldsymbol{m}_1,\boldsymbol{m}_2) \quad (1)$$

$$\varphi_M(\boldsymbol{m}_1,\boldsymbol{m}_2)=\varphi_{d_2}(\boldsymbol{m}_2)+\lambda_2\varphi_{\boldsymbol{m}_2}(\boldsymbol{m}_2)+\gamma_2\varphi_J(\boldsymbol{m}_1,\boldsymbol{m}_2) \quad (2)$$

where $\varphi_{d1}(\boldsymbol{m}_1)$ and $\varphi_{d2}(\boldsymbol{m}_2)$ are the misfits of AEM and AirMag data, while $\varphi_{m1}(\boldsymbol{m}_1)$ and $\varphi_{m2}(\boldsymbol{m}_2)$ are the constraints on the AEM and AirMag models, respectively. $\varphi_J(\boldsymbol{m}_1, \boldsymbol{m}_2)$ is the joint constraint formed by the PCCs. λ is the weighting coefficient for the model constraint $\varphi_m(\boldsymbol{m})$ in the objective function, representing the regularization factor, while γ is the weighting coefficient for the joint constraint $\varphi_J(\boldsymbol{m}_1, \boldsymbol{m}_2)$. Writing (1) and (2) in matrix format, we have

$$\varphi_{\mathrm{EM}}(\boldsymbol{m}_1,\boldsymbol{m}_2)=\left\|\boldsymbol{C}_{\boldsymbol{d}_1}\left(\boldsymbol{d}_1^{\mathrm{obs}}-\boldsymbol{F}(\boldsymbol{m}_1)\right)\right\|_2^2+\lambda_1\left\|\boldsymbol{C}_{\boldsymbol{m}}\left(\boldsymbol{m}_2-\boldsymbol{m}_2^{\mathrm{ref}}\right)\right\|_2^2+\gamma_1\sum_i^n r_i\left(1-p_i^2\right) \quad (3)$$

$$\varphi_M(\boldsymbol{m}_1,\boldsymbol{m}_2)=\left\|\boldsymbol{C}_{\boldsymbol{d}_2}\left(\boldsymbol{d}_2^{\mathrm{obs}}-G(\boldsymbol{m}_2)\right)\right\|_2^2+\lambda_2\left\|\boldsymbol{C}_{\boldsymbol{m}}\boldsymbol{W}\left(\boldsymbol{m}_2-\boldsymbol{m}_2^{\mathrm{ref}}\right)\right\|_2^2+\gamma_2\sum_i^n r_i\left(1-p_i^2\right) \quad (4)$$

where $\boldsymbol{d}_{1,2}^{\mathrm{obs}}$ denotes the observed data, and $\boldsymbol{m}^{\mathrm{ref}}$ is the reference model determined according to the prior information. $\boldsymbol{F}$ is the forward operator for the AEM method. Here, we use the 3-D finite-difference method by Liu and Yin [24] to calculate the frequency-domain AEM responses. $\boldsymbol{G}$ is a linear operator in the AirMag method, which represents the contributions from all discretized grids for the survey data. $\boldsymbol{C}_{d1}$ and $\boldsymbol{C}_{d2}$ are diagonal matrices, representing the estimation of the standard deviation of noisy data for AEM and AirMag, respectively. $\boldsymbol{C}_m$ is used to constraint the smoothness of the recovered model. p_i and r_i are, respectively, the PCC in the subdomain and the weights, whose calculation will be introduced in Section II-C. $\boldsymbol{W}$ is the sensitivity weighting matrix defined by

$$\boldsymbol{W}=\mathrm{diag}\left(\boldsymbol{G}^{\mathrm{T}}\boldsymbol{G}\right)^{1/2} \quad (5)$$

which is used to counteract the influence of the skin effect in magnetic inversion. Since it depends only on the sensitivity matrix, it can avoid the influence of other parameters on depth weighting [25]. When the optimization methods are used to minimize the objective function in (1)–(4), it is necessary to compute the gradient of the two objective functionals. Their expressions are

$$\boldsymbol{g}_{\mathrm{EM}}(\boldsymbol{m}_1)=\frac{\partial\varphi_{\mathrm{EM}}}{\partial\boldsymbol{m}_1}=-2\boldsymbol{J}_{\mathrm{EM}}^{\mathrm{T}}\boldsymbol{C}_{\boldsymbol{d}_1}(\boldsymbol{d}_1-\boldsymbol{F}(\boldsymbol{m}_1))+2\lambda_1\boldsymbol{C}_{\boldsymbol{m}_1}\left(\boldsymbol{m}_1-\boldsymbol{m}_1^{\mathrm{ref}}\right)+\gamma_1\frac{\partial\varphi_{\boldsymbol{J}}}{\partial\boldsymbol{m}_1} \quad (6)$$

$$\boldsymbol{g}_M(\boldsymbol{m}_2)=\frac{\partial\varphi_M}{\partial\boldsymbol{m}_2}=-2\boldsymbol{J}_M^{\mathrm{T}}\boldsymbol{C}_{\boldsymbol{d}_2}(\boldsymbol{d}_2-\boldsymbol{F}(\boldsymbol{m}_2))+2\lambda_2\boldsymbol{C}_{\boldsymbol{m}_2}\boldsymbol{W}\left(\boldsymbol{m}_2-\boldsymbol{m}_2^{\mathrm{ref}}\right)+\gamma_2\frac{\partial\varphi_{\boldsymbol{J}}}{\partial\boldsymbol{m}_2} \quad (7)$$

where $\boldsymbol{J}_{\mathrm{EM}}$ and $\boldsymbol{J}_M$ are, respectively, the Jacobian matrices for AEM and AirMag that represent the influence of model parameters on the data. For AirMag inversion, $\boldsymbol{J}_M$ is analytical

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so it can be directly substituted into (7) to calculate the first term on the right side. However, for AEM inversion, $\boldsymbol{J}_{\mathrm{EM}}$ needs to be calculated by adjoint forward modeling [26]. The second term on the right side in the equations (6) and (7) can be calculated analytically. For the joint constraints, we give a detailed introduction on its derivation in the following section.

B. Pearson Correlation Coefficient

The PCC is a statistical method to judge the linear correlation between two types of data. PCC is not affected by the change in the position or scale of variables, and its value depends only on the linear correlation between the two types of data. Thus, the PCC is suitable for the evaluation of the linear correlation between data in different orders and dimensions [21]. We assume that there are two types of data, $\boldsymbol{X} = \{x_1, x_2, \ldots, x_n\}$, $\boldsymbol{Y} = \{y_1, y_2, \ldots, y_n\}$, with the same length but independent on each other, then the expression of PCC for them can be written as

$$p = \frac{\operatorname{cov}(\boldsymbol{X}, \boldsymbol{Y})}{\sigma_X \sigma_Y} = \frac{E(\boldsymbol{XY}) - E(\boldsymbol{X})E(\boldsymbol{Y})}{\sqrt{E\left(\boldsymbol{X}^2\right) - E^2(\boldsymbol{X})}\sqrt{E\left(\boldsymbol{Y}^2\right) - E^2(\boldsymbol{Y})}} = \frac{\sum (x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum (x_i - \overline{x})^2}\sqrt{\sum (y_i - \overline{y})^2}} \tag{8}$$

where $\operatorname{cov}(\boldsymbol{X}, \boldsymbol{Y})$ represents the covariance of data $\boldsymbol{X}$ and $\boldsymbol{Y}$, σ_X, σ_Y is the standard deviation of the corresponding data, $\overline{x} = E(\boldsymbol{X})$ and $\overline{y} = E(\boldsymbol{Y})$ are the expected values of data $\boldsymbol{X}$ and $\boldsymbol{Y}$, respectively. The value of PCC ranges from -1 to 1, which describes the trend of synchronous change of the two types of data. When the linear correlation of the two variables increases, the correlation coefficient tends to 1 or -1. When the value of PCC is positive, it means a positive correlation, namely, the two data increase or decrease simultaneously. On the contrary, a negative value means a negative correlation, so the two data vary in opposite directions. A correlation coefficient that is close to 0 indicates that no linear correlation between the two data exists. Generally, when $|p| > 0.8$, it is believed that the linear correlation is strong.

C. LPCC Computation

For a single target body with multiple properties, there must be a certain correlation between different properties that can be used in the joint inversion. In this case, the correlation is usually approximately linear that can be represented by PCC. However, when there are multiple abnormal bodies in the underground, even if the correlation between them is strong, the total correlation in the whole area may not be linear. As a result, the correlation between different geophysical parameters in the entire survey area cannot be well defined by PCC. To overcome this problem, we propose a local PCC method that separates the entire domain into subdomains of small size. By choosing the size appropriately, the correlation in the subdomain becomes approximately linear that can be well represented by PCC. In geophysical exploration, although the correlation of different physical parameters in the entire area is not strong, they may have a strong PCC in local areas, which lays the foundation for joint inversion.

Here, we use a 2-D model as an example to show the property of local PCC. Refer to Fig. 1(a) and (b), we select three different zones as subdomains to calculate the local PCCs and compare them with the PCC for the entire model. We convert the 2-D models into 1-D vector to show the distribution patterns of resistivity and magnetization. From Fig. 1(c)–(e), one may observe that the correlation for the entire model and zone 2 are not strong, but there exists a strong linear correlation ($|p| > 0.8$) in the zone 1 and 3. The PCC difference in three subzones indicates that the PCC can well represent the correlation of different parameters in local areas. This further proves that the PCC can be a suitable structural similarity constraint for joint inversions. Based on this, we take the LPCC-based constraint as the regularization term in (3) and (4). To do that, we take the current grid as the center, and subtract those grids around it but within a certain range to form a 3-D subdomain. This operation loops over all grids. Meanwhile, it is believed that the geophysical parameters in each subdomain meet the approximately linear condition. To calculate the PCC for subdomain i that contains M_i units, we define $\boldsymbol{m}_{1i}^j$ and $\boldsymbol{m}_{2i}^j$ as the two model parameters of the jth grid in subdomain i. $\overline{\boldsymbol{m}}_{1i}$ and $\overline{\boldsymbol{m}}_{2i}$ with length M_i are defined as the average values of the two parameters. According to the PCC definition in (8), we have for a subdomain

$$p_i = \frac{\sum_j^{M_i}\left(\boldsymbol{m}_{1i}^j - \overline{\boldsymbol{m}}_{1i}\right)\left(\boldsymbol{m}_{2i}^j - \overline{\boldsymbol{m}}_{2i}\right)}{\sqrt{\sum_j^{M_i}\left(\boldsymbol{m}_{1i}^j - \overline{\boldsymbol{m}}_{1i}\right)^2}\sqrt{\sum_j^{M_i}\left(\boldsymbol{m}_{2i}^j - \overline{\boldsymbol{m}}_{2i}\right)^2}} = \frac{\widetilde{\boldsymbol{m}}_{1i}^{\mathrm{T}} \cdot \widetilde{\boldsymbol{m}}_{2i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2 \|\widetilde{\boldsymbol{m}}_{2i}\|_2} \tag{9}$$

where $\widetilde{\boldsymbol{m}}_{1i} = (\ldots, \boldsymbol{m}_{1i}^j - \overline{\boldsymbol{m}}_{1i}, \ldots)^{\mathrm{T}}$, $\widetilde{\boldsymbol{m}}_{2i} = (\ldots, \boldsymbol{m}_{2i}^j - \overline{\boldsymbol{m}}_{2i}, \ldots)^{\mathrm{T}}$. Each element of these two vectors represents the difference between the model parameters of jth grids and the average values of the subdomain.

When the fluctuations of the two datasets are small, the two terms in the denominator of (9) are very small, so the PCC in (9) will appear singular. To solve this problem, we introduce a weighting factor r_i in above (3) and (4) as

$$r_i = \begin{cases} 1, & \sigma_i^{m_1} > \varepsilon_1 \text{ and } \sigma_i^{m_2} > \varepsilon_2 \\ 0, & \sigma_i^{m_1} \le \varepsilon_1 \text{ or } \sigma_i^{m_2} \le \varepsilon_2 \end{cases} \tag{10}$$

where σ_i^{m1} and σ_i^{m2}, respectively, represent the standard deviation of two model variations in the subdomain, and ε_1, ε_2 are the thresholds, employed to avoid the meaningless constraints caused by too small σ_i^{m1} or σ_i^{m2}. When one or both of σ_i^{m1} and σ_i^{m2} are less than the predefined threshold, r_i is equal to 0; otherwise, r_i is equal to 1.

The threshold value can be defined as the average value of the standard deviation of all model variations, i.e.,

$$\varepsilon_k = \sum_{i=1}^{M} \sigma_i^{m_k} / M \tag{11}$$

where $k = 1$ or 2, representing the two parameters. M is the total number of subdomains. Based on (11), we can specifically constrain the correlation in the area above the average standard deviation of the model parameters. The ε_1

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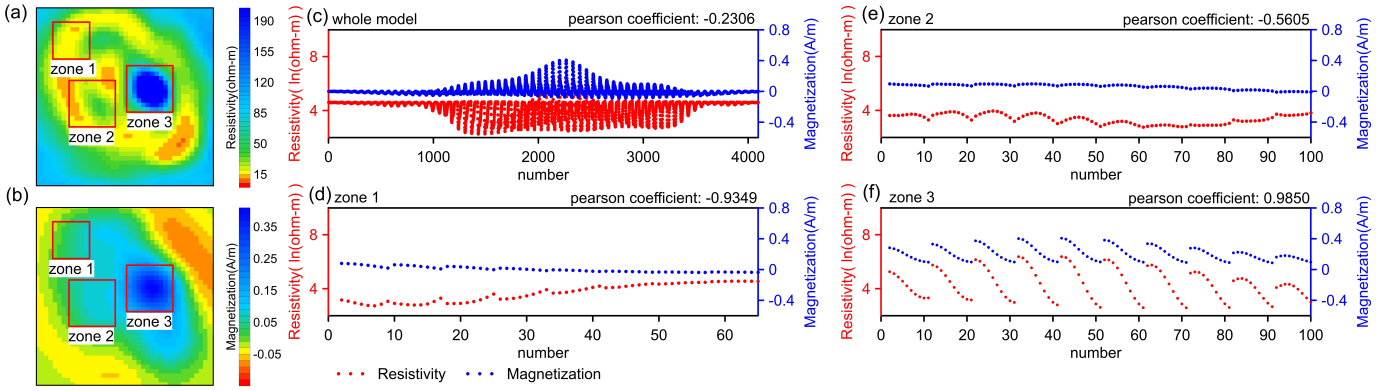

Fig. 1. Calculation diagram of PCC: (a) and (b) 2-D model of resistivity and magnetization and (c)–(f) rearranged 1-D array for resistivity and magnetization for the entire model and subdomains of zone 1, zone 2, and zone 3, respectively.

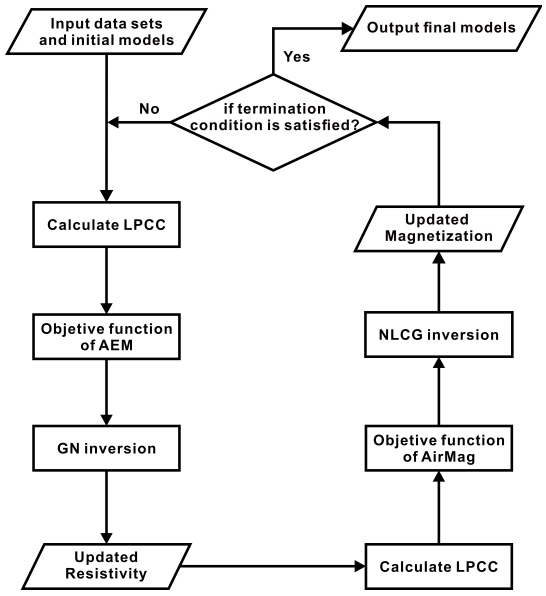

Fig. 2. Workflow of joint inversion based on LPCC.

and ε_2 in (11) vary continuously in the inversion to ensure that the constrained zones are consistent with large model variations. This means that when the LPCC constraint terms in (3) and (4), we can have a good identification of the anomalous boundary. Moreover, from the last terms in (3) and (4), we can see that when the correlation in a subdomain is stronger, p_i^2 is closer to 1, otherwise, p_i^2 tends to 0. As the inversion converges, the LPCC term $1-p_i^2$ for each subdomain becomes smaller and smaller, and the correlation becomes stronger and stronger, to finally achieve the goal of joint inversion. From the above discussion, we can summarize the characteristics of LPCC as follows.

1) When the two model variances are larger than the thresholds in the same zone, if they change in the same direction or in the opposite direction with a similar distribution pattern, the value of LPCC is close to 0; Otherwise, if the changing pattern is not consistent, LPCC has big value.
2) If one or two of the model variances are smaller than the threshold, the value of LPCC is 0. This allows only one model to vary without being affected by the other.
3) The LPCC can be calculated in a multiscale subdomain, thus we can use it to obtain aimed model features by doing joint inversion in different scales.
4) In the LPCC computation, we do not need to calculate the gradients of model parameters in orthogonal directions. This makes it easier to be used in unstructured grids.

D. Gradient of Objective Function

To obtain the gradient in (3), we need to calculate the derivatives of LPCC regularization term $\varphi_J(\boldsymbol{m}_1, \boldsymbol{m}_2)$ in (6) with respect to (w.r.t.) $\boldsymbol{m}_1$, that is,

$$\frac{\partial \varphi_J}{\partial \boldsymbol{m}_1} = -2\sum_{i=1}^{N} r_i p_i \frac{\partial p_i}{\partial \boldsymbol{m}_1}. \tag{12}$$

Equation (12) is equivalent to calculating the sum of the derivatives in each subdomain w.r.t. $\boldsymbol{m}_1$. The key step is to calculate the derivative of p_i. Substituting (9) into equation (12), we have

$$\begin{aligned}\frac{\partial p_i}{\partial \boldsymbol{m}_1} &= \frac{\partial}{\partial \boldsymbol{m}_1}\left(\frac{\widetilde{\boldsymbol{m}}_{1i}^{\mathrm{T}}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2}\right)\cdot\frac{\widetilde{\boldsymbol{m}}_{2i}}{\|\widetilde{\boldsymbol{m}}_{2i}\|_2} \\ &= \frac{\partial \widetilde{\boldsymbol{m}}_{1i}^{\mathrm{T}}}{\partial \boldsymbol{m}_1}\cdot\left(\frac{\widetilde{\boldsymbol{m}}_{2i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2\|\widetilde{\boldsymbol{m}}_{2i}\|_2} - p_i\frac{\widetilde{\boldsymbol{m}}_{1i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2^2}\right).\end{aligned} \tag{13}$$

Here, $\partial p_i/\partial \boldsymbol{m}_1 = (\partial p_i/\partial \boldsymbol{m}_{1i}^{j}, \ldots, \partial p_i/\partial \boldsymbol{m}_{1i}^{M_i})^{\mathrm{T}}$ with j denoting the index of the grid in the total M_i grids in the ith subdomain. For grids outside the current subdomain $\partial \widetilde{\boldsymbol{m}}_{1i}/\partial \boldsymbol{m}_1 = 0$. For a grid within the current subdomain, the element $\partial \widetilde{\boldsymbol{m}}_{1i}/\partial \boldsymbol{m}_1$ is

$$\frac{\partial \widetilde{\boldsymbol{m}}_{1i}^{j}}{\partial \boldsymbol{m}_{1i}^{k}} = \begin{cases} 1-\dfrac{1}{M_i}, & j=k \\ -\dfrac{1}{M_i}, & j\neq k \end{cases} \quad j,k \le M_i \tag{14}$$

where k and j represent the index of the grid in the subdomain. Moreover, substituting (13) and (14) into (12), we obtain

$$\frac{\partial \varphi_J}{\partial \boldsymbol{m}_1} = -2\sum_{i=1}^{M} r_i p_i \frac{\partial \widetilde{\boldsymbol{m}}_{1i}^{\mathrm{T}}}{\partial \boldsymbol{m}_1}\cdot\left(\frac{\widetilde{\boldsymbol{m}}_{2i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2\|\widetilde{\boldsymbol{m}}_{2i}\|_2} - p_i\frac{\widetilde{\boldsymbol{m}}_{1i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2^2}\right). \tag{15}$$

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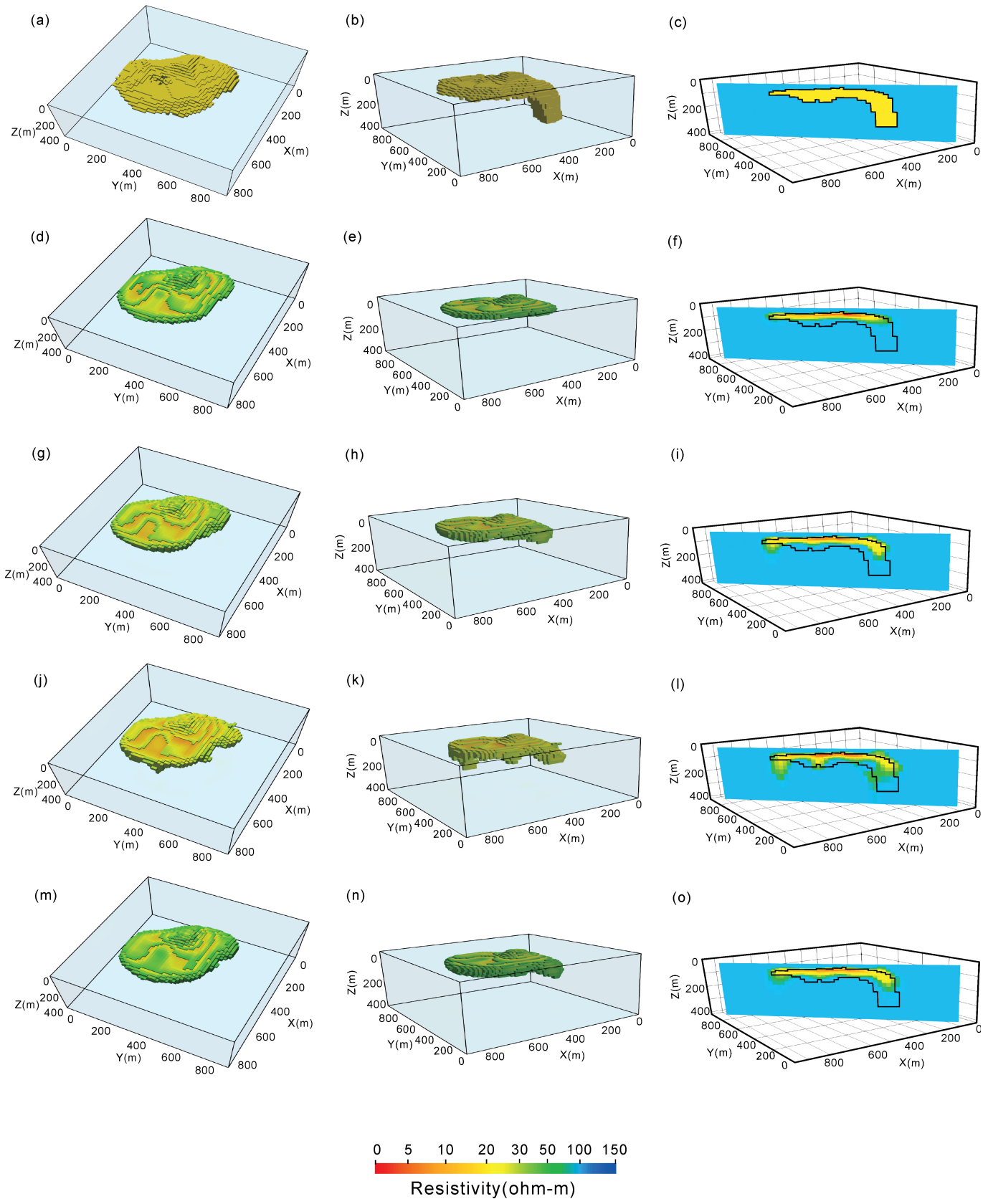

Fig. 3. True resistivity model and results of individual and joint inversion of AEM and AirMag data: (a)–(c) true resistivity models observed from different directions; (d)–(f) individual AEM inversion results; (g)–(i) cross-gradient joint inversion results; (j)–(l) joint inversion results with 8 × 8 × 6 subdomain; and (m)–(o) joint inversion results with 8 × 8 × 10 subdomain. The black lines in (c), (f), (i), (l), and (o) mark the anomalous boundary in the vertical slice.

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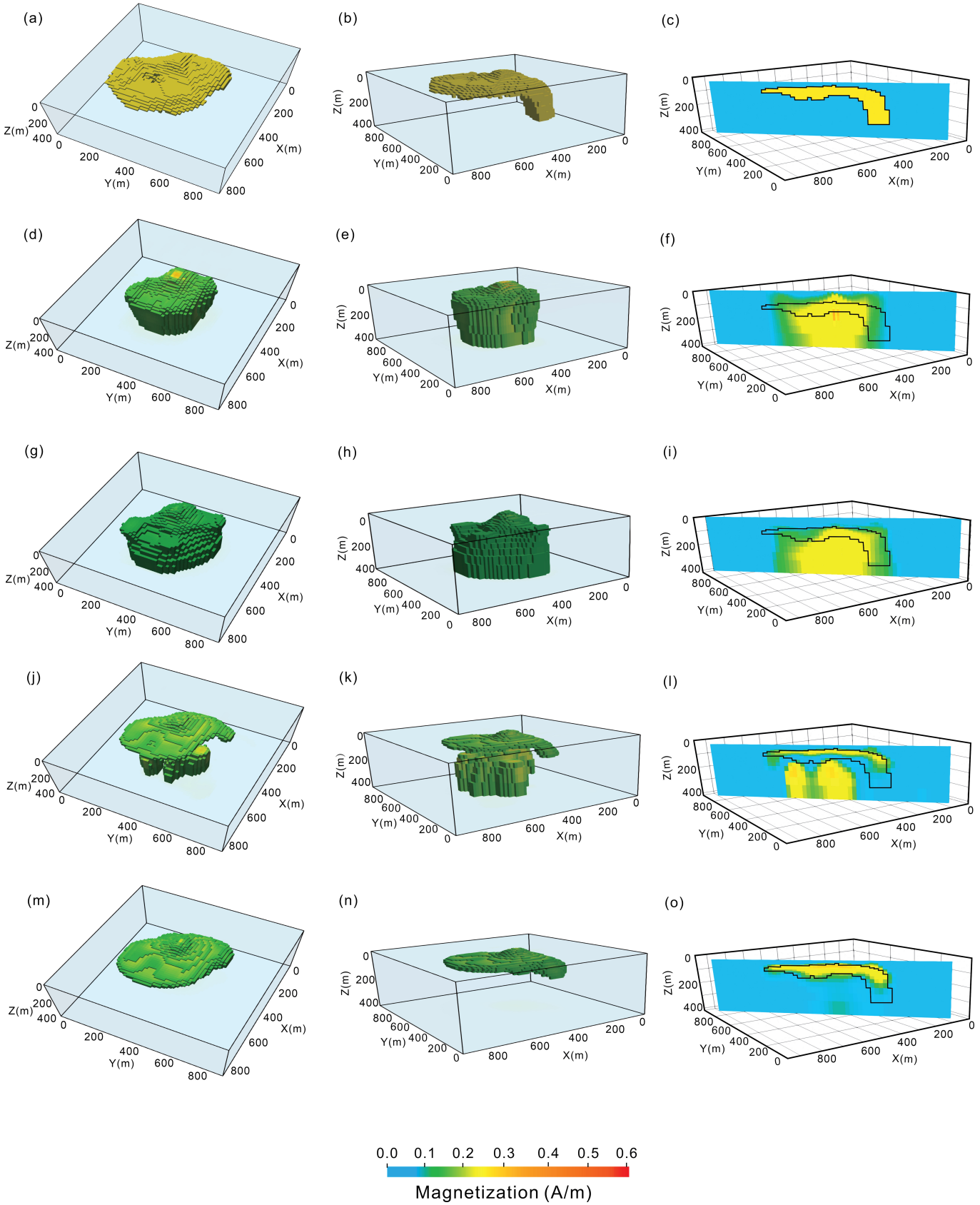

Fig. 4. True resistivity model and results of individual and joint inversion of AEM and AirMag data: (a)–(c) true resistivity models observed from different directions; (d)–(f) individual AirMag inversion results; (g)–(i) cross-gradient joint inversion results; (j)–(l) joint inversion results with $8 \times 8 \times 6$ subdomain; and (m)–(o) joint inversion results with $8 \times 8 \times 10$ subdomain. The black lines in (c), (f), (i), (l), and (o) mark the anomalous boundary in the vertical slice.

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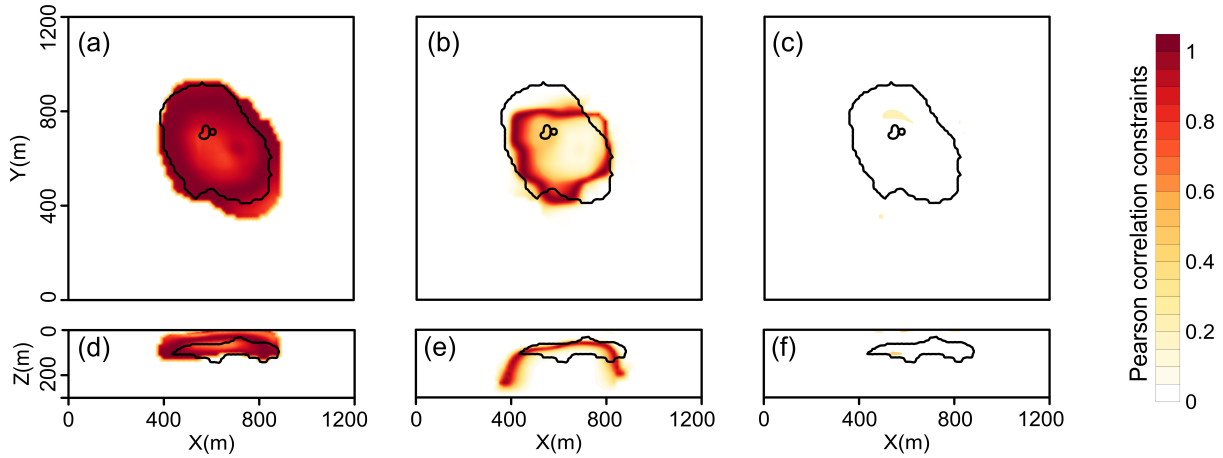

Fig. 5. Pearson correlation constraints for individual and LPCC joint inversions: (a)–(c) horizontal slices at $z = 90$ m for individual inversions and joint inversions with $8 \times 8 \times 6$ and $8 \times 8 \times 10$ subdomains and (d)–(f) vertical slices at $y = 600$ m of the corresponding inversion results. The black lines mark the anomalous boundary.

Equation (15) gives the derivative of LPCC regularization term $\varphi_J(\boldsymbol{m}_1, \boldsymbol{m}_2)$ w.r.t. model $\boldsymbol{m}_1$. The derivative w.r.t. model $\boldsymbol{m}_2$ can be obtained similarly, i.e.,

$$\frac{\partial \varphi_J}{\partial \boldsymbol{m}_2} = -2\sum_{i=1}^{M} r_i p_i \frac{\partial \widetilde{\boldsymbol{m}}_{2i}^{\mathrm{T}}}{\partial \boldsymbol{m}_2} \cdot \left(\frac{\widetilde{\boldsymbol{m}}_{1i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2 \|\widetilde{\boldsymbol{m}}_{2i}\|_2} - p_i \frac{\widetilde{\boldsymbol{m}}_{2i}}{\|\widetilde{\boldsymbol{m}}_{2i}\|_2^2} \right). \tag{16}$$

According to (15) and (16), the gradients $\boldsymbol{g}_{\mathrm{EM}}$ and $\boldsymbol{g}_M$ of objective functions in (6) and (7) are

$$\begin{aligned} \boldsymbol{g}_{\mathrm{EM}} = & -2\boldsymbol{J}_{\mathrm{EM}}^{\mathrm{T}} \boldsymbol{C}_{\boldsymbol{d}_1} \Delta \boldsymbol{d} + 2\lambda \boldsymbol{C}_m \left(\boldsymbol{m}_1 - \boldsymbol{m}_1^{\mathrm{ref}} \right) \\ & -2\sum_{i=1}^{M} r_i p_i \frac{\partial \widetilde{\boldsymbol{m}}_{1i}^{\mathrm{T}}}{\partial \boldsymbol{m}_1} \cdot \left(\frac{\widetilde{\boldsymbol{m}}_{2i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2 \|\widetilde{\boldsymbol{m}}_{2i}\|_2} - p_i \frac{\widetilde{\boldsymbol{m}}_{1i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2^2} \right) \end{aligned} \tag{17}$$

$$\begin{aligned} \boldsymbol{g}_M = & -2\boldsymbol{J}_M^{\mathrm{T}} \boldsymbol{C}_{\boldsymbol{d}_2} + 2\lambda \boldsymbol{C}_m \boldsymbol{W} \left(\boldsymbol{m}_2 - \boldsymbol{m}_2^{\mathrm{ref}} \right) \\ & -2\sum_{i=1}^{M} r_i p_i \frac{\partial \widetilde{\boldsymbol{m}}_{2i}^{\mathrm{T}}}{\partial \boldsymbol{m}_2} \cdot \left(\frac{\widetilde{\boldsymbol{m}}_{1i}}{\|\widetilde{\boldsymbol{m}}_{1i}\|_2 \|\widetilde{\boldsymbol{m}}_{2i}\|_2} - p_i \frac{\widetilde{\boldsymbol{m}}_{2i}}{\|\widetilde{\boldsymbol{m}}_{2i}\|_2^2} \right) \end{aligned} \tag{18}$$

where $\Delta \boldsymbol{d}$ represents the difference between the observed data and the model results.

E. Workflow of Joint Inversion

The conventional joint inversions strategy unifies the data fitting term, model fitting term, and joint constraint term into one objective function. These methods need to set weights for each dataset joining the inversion, which can impact the data fitting in the inversion process. To avoid this problem, we adopt the alternative strategy proposed by Gao and Zhang [22] and use (1) and (2) for our joint inversions. Fig. 2 shows the procedure for our joint inversion. In the optimization of the objective function, the two inversions for different data are carried out independently, but they are connected by an LPCC term as the joint constraint. We use the Gauss–Newton method in our AEM inversion, where the conjugate gradient method is used to solve the linear system of equations to obtain the model updates. For the AirMag inversion, we adopt the nonlinear conjugate gradient (NLCG) algorithm with the descent step searched by linear search technique [27].

After one model is updated, the LPCC term is updated simultaneously and substituted into the objective function for another dataset. In this way, the two datasets are inverted alternately until the condition for the joint inversions is met. When (1) and (2) are used as the objective functions, the weight coefficients λ and γ need to be constantly updated in the inversion.

In AEM and AirMag inversions, the root mean square (rms) of data misfit is taken as one of the conditions for terminating the iterations. If the rms approaches 1.0 for AEM inversion or is smaller than 0.1 for AirMag inversion, we believe that the data are well fitted, so the inversions will be terminated. For cases when the rms cannot reach the predefined threshold, especially for the field data inversion, we set another terminating criterion, namely when the rms drops too slowly for the smallest tradeoff parameter λ, the inversion stops.

For the tradeoff parameter λ_1, we select a large value at the beginning of inversion to provide a strong model constraint, and we reduce it when the rms drops slowly until it reaches the threshold. However, for parameter λ_2, we set a fixed value by many tests. This is because the AirMag data are easy to fit so we do not need to change the parameter to improve the data fitting. The way of selecting joint constraint weights γ is different from that for λ. At the beginning of the inversion, the data fitting is weak. To avoid too strong constraints from the joint term, we give a small γ initially. With the data fitting being improved, we increase γ to an appropriate level to enhance the correlation constraint on the two models. The updating formula is $\gamma^{i+1} = \lambda^i \varphi_m^i / \varphi_j^i$. That is, the joint constraint term should be equal to the weighted model constraint. At the final stage of inversion, γ is reduced to improve the data fitting.

III. Numerical Experiments

To test the effectiveness of the proposed joint inversion algorithm, we design a synthetic model, as shown in Fig. 3(a)–(c).

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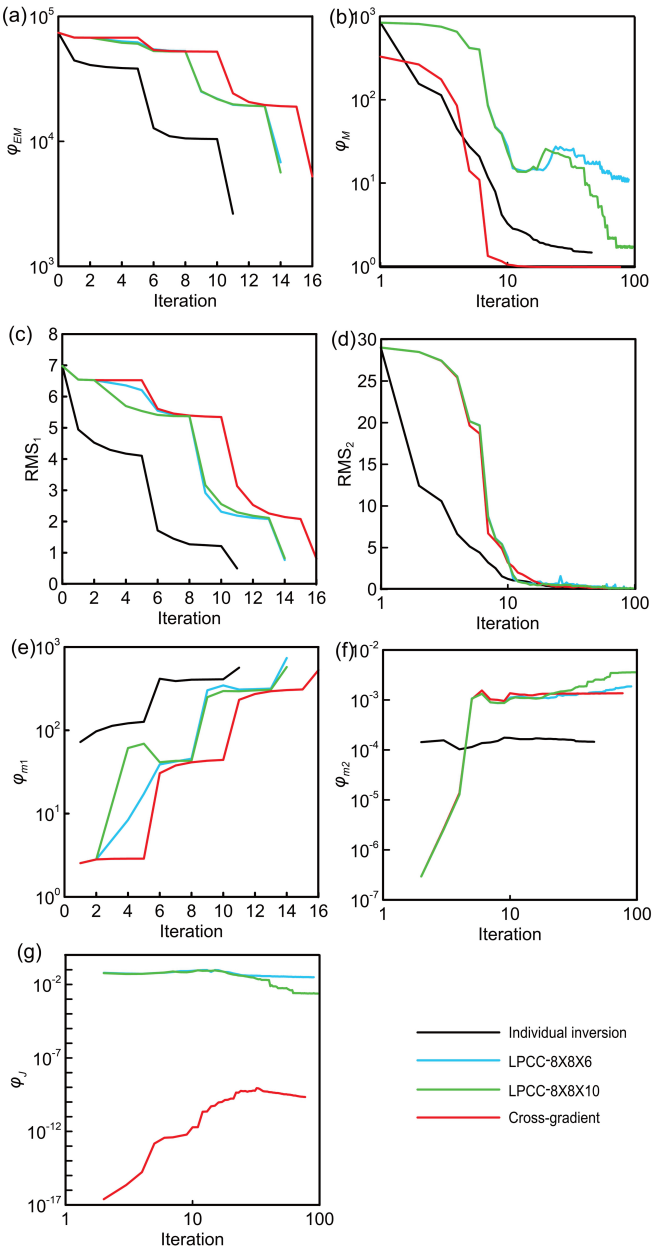

Fig. 6. (a) Objective functional; (b) rms; and (c) model regularization terms for AEM inversion; (d) objective functional, (e) rms, and (f) model regularization terms for AirMag inversion; and (g) regularization terms for the joint inversion.

The abnormal body locates in the center of the survey area. Its top depth is 50 m, extending downward to about 350 m. We use the same mesh to invert both AEM and AirMag data. The entire inversion model is discretized into 64 × 64 × 38 grids with eight air layers.

There is a total of 128 × 128 × 30 cells in our forward modeling with the horizontal sizes in both x- and y-direction being 7.5 m. However, in the inversion, we discretize the model into 64 × 64 × 30 cells and the grid sizes in the x- and y-direction are both 15 m. In both cases, we extract for each survey point the footprint size (200 m × 200 m) to construct the computational model and add four expanded grids in each direction with the sizes of 20, 40, 80, and 160 m. The vertical size of the first grid in the underground for the inversion is

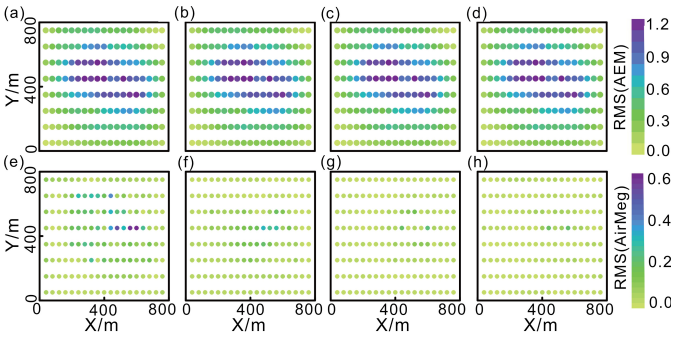

Fig. 7. Data fitting at each survey point for: (a) and (e) individual inversions; (b) and (f) cross-gradient joint inversions; (c) and (g) LPCC joint inversion with 8 × 8 × 6 subdomain; and (d) and (h) LPCC joint inversions with 8 × 8 × 10 subdomain.

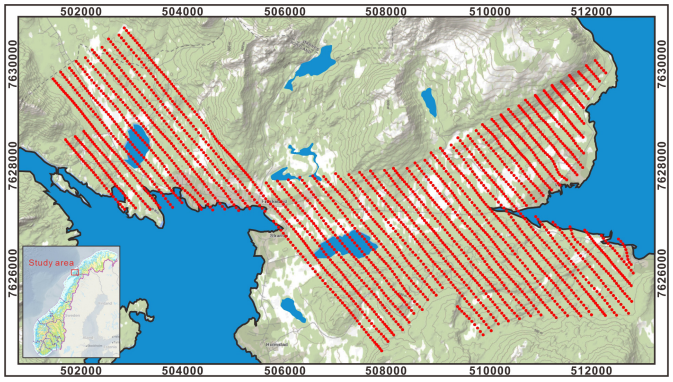

Fig. 8. Survey lines and stations in Lofoten–Vesteralen, Northern Norway. The red dots mark the survey stations. (Modified from [28].)

2 m, while those of the following grids are calculated by continuously multiplying the vertical size of the above grid by a factor of 1.1. This process loops until the 26th grid. The vertical sizes of the final four grids are set to be 100, 200, 400, and 600 m, respectively. The overall thickness of the entire model in the vertical direction is 1617 m. The resistivity of the abnormal body is 10 Ω·m, and the resistivity of the background is 100 Ω · m. For the magnetic method, the magnetization of the abnormal body is set to be 0.3 A/m, while the background magnetization is 0 A/m.

We design eight survey lines in the area with a line spacing of 100 m. There are 19 survey points on each line at a 40-m interval. Thus, totally we have 152 survey points. The airborne EM system used is a frequency-domain helicopter system from the Norwegian Geological Survey with five frequencies of 880, 6606, and 7001 Hz for horizontal coplanar (HCP) coil array, 980, and 34 133 Hz for vertical coaxial (VCA) coil array. We add 3% Gaussian noise to AEM and AirMag data to simulate the field situation. All examples in this article are calculated on Dell workstation with 8 core Xeon CPU, 128-GB RAM, under Windows 10 platform. The time costs for individual AEM and AirMag inversion are, respectively, about 6.5 and 0.1 h, while the joint inversion takes about 7 h.

In order to analyze the reliability of the algorithm and the influence of subdomain sizes on the joint inversions, we have done the cross-gradient joint inversion and set two subdomain cases of 8 × 8 × 6 and 8 × 8 × 10 grids. The LPCC-based joint inversion is applied to the two cases

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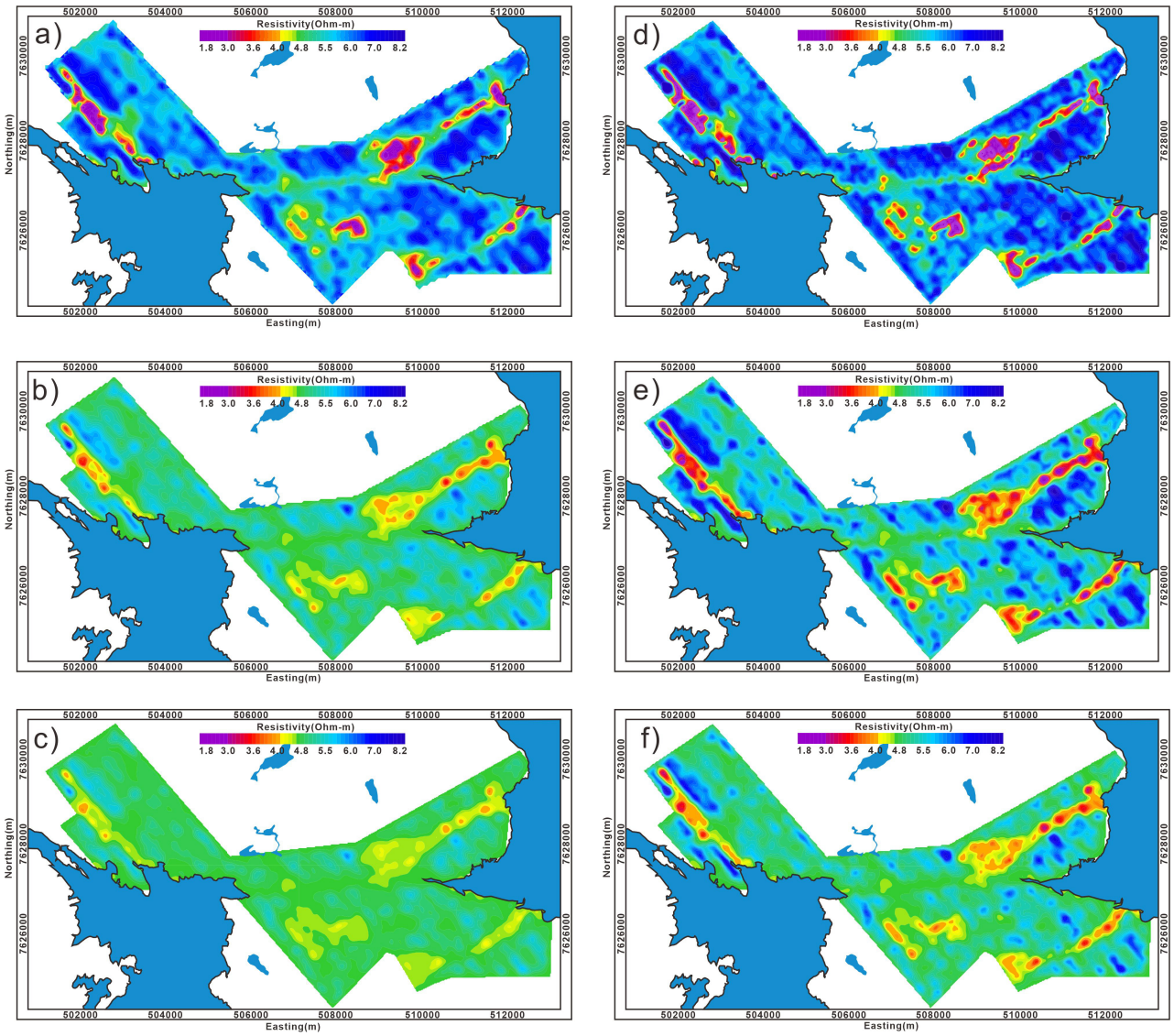

Fig. 9. Inversion results of the field survey data acquired from Lofoten–Vesteralen, Northern Norway: (a)–(c) are results of individual AEM inversion at $z = 40$, 90, and 140 m, while (d)–(f) are results of LPCC joint inversions at the same depths.

with the same termination conditions, model constraints, and joint constraints. As shown in Fig. 3(d)–(f), the individual AEM inversion can well recover the horizontal location of the anomalous body, and the resistivity distribution is smoothed as we use an L_2-norm-based smoothing regularization. However, due to the high frequencies used in the AEM system, the skin depth is small, thus the AEM inversion cannot well recover the thickness of the target and the bottom shape. In the results of joint inversion based on the cross-gradient and LPCC, however, the obtained resistivity models show better shape at the bottom of the anomalous body. Especially from the right panel in Fig. 3, we can see a downward-extending part, which is consistent with the true model. It should be noted that when the subdomain is $8 \times 8 \times 6$, the results of joint inversion using LPCC have false anomalies below the anomaly body. This is because, in the two individual inversions of AEM and AirMag data [Figs. 3(d)–(f) and 4(d)–(f)], the recovered resistivity and magnetization model have small overlays in the vertical direction. If the size of the subdomain is small, the local correlation only affects a few grids, resulting in the LPCC being unable to well constrain two individual inversions. As the size of the subdomain increases, the two recovered anomalous bodies effectively constrain each other, so the final results are improved.

From Fig. 4(d)–(f), we can see that the individual AirMag inversion can only roughly determine the horizontal location of the anomalous body, but in the vertical direction, the resolution is very poor. This is because the AirMag inversion is a linear inversion, where the data can easily be fitted, while the solution can be seriously nonunique. The depth weighting matrix $\boldsymbol{W}$ in (5) can improve slightly the vertical resolution, but cannot help much in the optimization of AirMag inversions. From the joint inversions, we can see that the LPCC effectively improves the horizontal and vertical resolution of the magnetization model when the subdomain size is appropriate. In contrast, the cross-gradient method only improves the shape of the anomalous body at the top, while the improvement to the bottom of the anomalous body is weak [see Fig. 4(g)–(i)].

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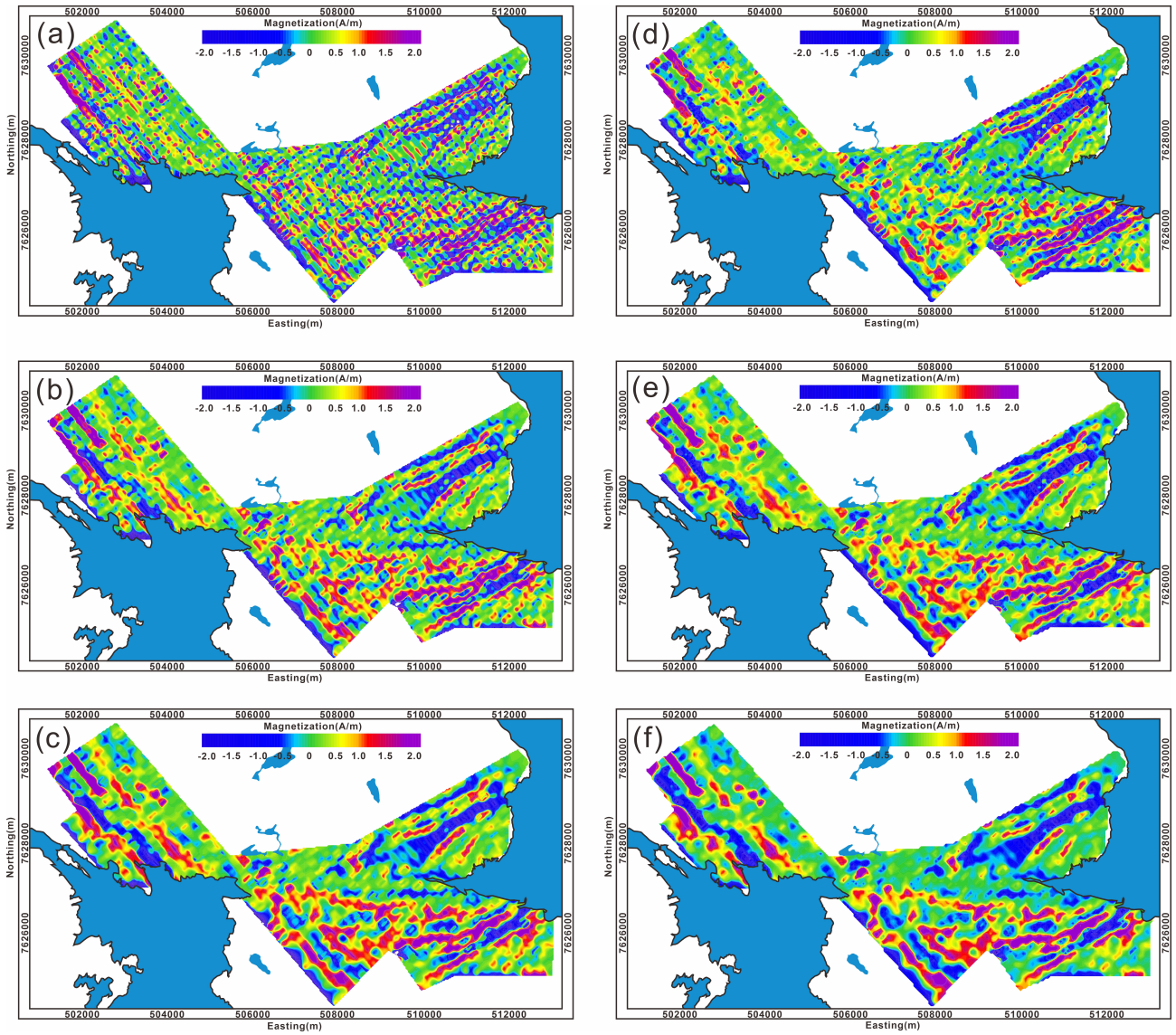

Fig. 10. Inversion results of the field survey data acquired from Lofoten–Vesteralen, Northern Norway: (a)–(c) are results of individual AirMag inversions at $z = 40$, 90, and 140 m, while (d)–(f) are results of LPCC joint inversions at the same depths.

This is because the cross-gradient constraint is done within the neighbor grids, it can be taken as a "local constraint," which can well constrain the structures on local scale. However, in the individual inversions of this example, the anomalous bodies recovered by AEM and AirMag only overlap at the top, while the deep structures are quite different. This results in hardship for the cross-gradient method to modify the bottom structure. For the LPCC inversion, when the size of the subdomain is small, it works similar to the cross-gradient method [as shown in Fig. 4(j)–(l)]; when the size of the subdomain becomes larger, it can control the anomalous structure on larger scales, thus the results in Fig. 4(m)–(o) are much better.

Fig. 5 shows the slices of LPCC obtained by the individual and joint inversions. As we know, if the LPCC value is close to 0, the correlation between AEM and AirMag anomalies is good. From Fig. 5, we can see that the correlations of the joint inversions are much better than the individual ones. For the $8 \times 8 \times 6$ subdomain, the inversion does not well constrain between the resistivity and magnetization anomalies, so the correlation does not improve very much. However, for the subdomain $8 \times 8 \times 10$, the LPCC becomes very small, so the correlation between AEM and AirMag model is very much improved. This implies that the appropriate subdomain size is important for the LPCC-based joint inversion. At present, the subdomain size is selected based on experience and the individual inversion results. If the inversion results are different on large scale, we need to choose a large subdomain; otherwise, we use a small subdomain to capture structural details. In general, the size of the subdomain in three directions should be bigger than 2. Meanwhile, the subdomain cannot be too large, as it cannot satisfy the assumption of linear correlation, or else the details in the inversions will get lost.

Fig. 6 shows the objective functional, rms, the model regularization terms, and the joint regularization term versus iterations. It is observed that the joint inversions are a little slower than the individual ones. This is because the mutual constraints between the AEM and AirMag anomalies slow down the inversion process. However, in terms of final results,

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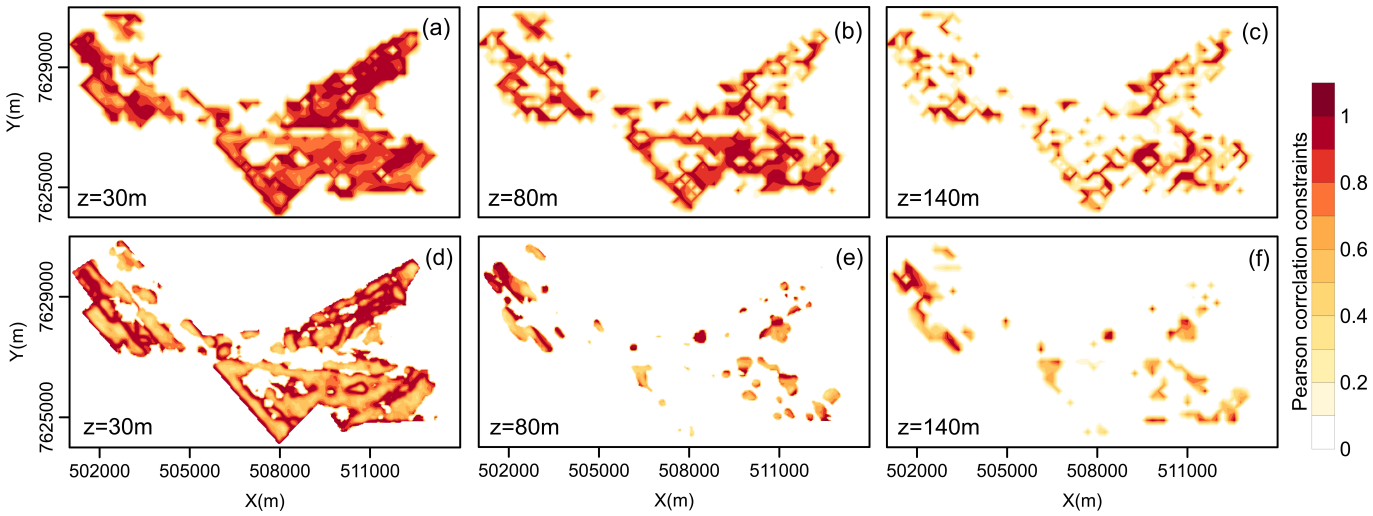

Fig. 11. Pearson correlation constraints for individual and LPCC joint inversions at different depths: (a)–(c) horizontal slices at $z = 30$, 80, and 140 m for individual inversions and (d)–(f) horizontal slices at $z = 30$, 80, and 140 m for LPCC joint inversion with $8 \times 8 \times 10$ subdomains.

our inversions meet the convergence criteria and the data fitting is good. In Fig. 7, we also show the data fitting at each survey point. It is observed that while not much improvement can be seen from the AEM data fitting, the AirMag data fitting in the joint inversion is better than that in the individual inversions. This indicates that the joint inversion improves the inversion result.

IV. Inversion of Field Data

In order to further test the effectiveness of our LPCC-based joint inversion algorithm, we invert the AEM and AirMag field data acquired in the Lofoten–Vesteralen area, Norway. The AEM system is the same as that used in synthetic inversions. Fig. 8 shows the AEM and AirMag survey lines and stations [28].

The entire inversion model is discretized into $608 \times 308 \times 25$ grids with eight air layers. The size of grids in x- and y-directions is 30 m. The overall extension of the model is 19 800 m in the x-direction and 10 800 m in the y-direction. The vertical size of the first grid in the underground is 6 m, and the vertical sizes of the following grids are increased by continuously multiplying the vertical size of the above grid by a factor of 1.2 until the 13th grid. The vertical sizes of the final four grids are set to be 100, 200, 400, and 600 m. The four grids at the bottom increase by a factor of 2 as the expanded grid. The total thickness of the model is 3000 m. The subdomain size of joint inversion is set to be $8 \times 8 \times 10$. The initial model is a uniform half-space with a resistivity of 100 $\Omega \cdot$ m and magnetization of 0 A/m. The individual AEM and AirMag inversion takes, respectively, about 97 and 0.4 h to converge, while the joint inversion takes 101 h.

Figs. 9 and 10 show the inversion results. From Fig. 9, it can be observed that the joint inversion based on LPCC obtains better horizontal resolution than the individual AEM inversion. When the depth reaches 140 m, the AEM inversion cannot well recover the anomalous location and resistivity, but the joint inversion provides more accurate results. According to the NGU report by Rønning *et al.* [28], the graphite ore

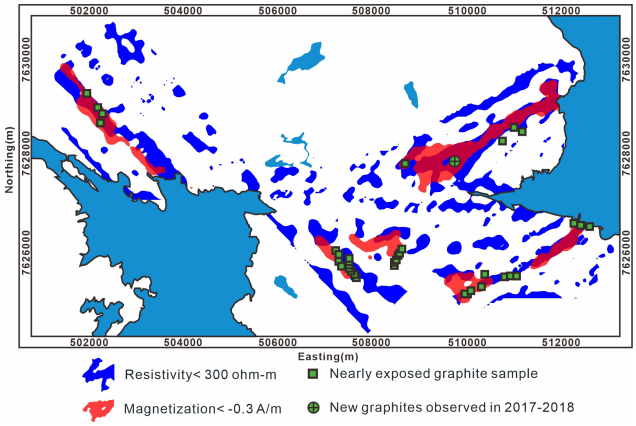

Fig. 12. Anomalies of low resistivity and low magnetization from the LPCC joint inversions and the distribution of graphite ore.

body extends to a large depth, thus we think the joint inversion result is more reliable. From Fig. 10, it can be observed that the results individual AirMag inversions are seriously affected by the location of the survey line, especially in the near-surface, demonstrating the strip pattern of anomalies. This is because the AirMag data inversion has strong nonuniqueness, the field data can be fitted in many ways. Since the grids close to the survey stations have large sensitivity, the AirMag inversion tries to fit the data by varying parameters close to these grids, so that the inversion results show many separated linear features. From the joint inversions in Fig. 10(d) and (e), these linear features are smeared out, so the anomalies become horizontally more continuous. From these results, we draw the conclusion that by combining the nonlinear character of AEM inversion and using LPCC in the joint inversion, the AirMag inversion delivers improved results. Fig. 11 shows the distribution of LPCC for individual and joint inversions at different depths. It can be observed that the joint inversions enforce the correlation of the resistivity and magnetization model in the deep earth.

In Fig. 12, we project the joint inversion results of resistivity and magnetization on the distribution map of graphite ore in this area. The green marks denote the graphite outcrops. It can

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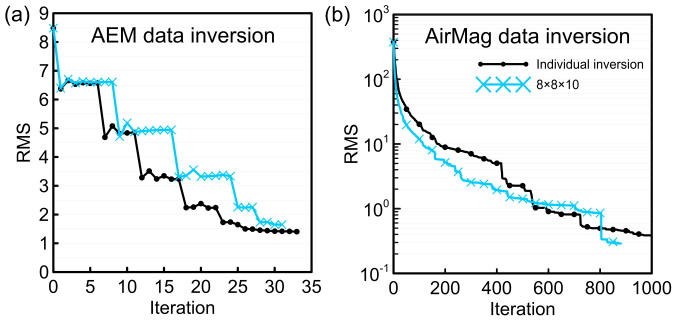

Fig. 13. Data misfit (rms) of individual and LPCC joint inversions for the survey data from Lofoten–Vesteralen, Northern Norway. (a) AEM data. (b) AirMag data.

be observed that the areas where the anomalies from AEM and AirMag inversions are overlapped contain more likely graphite ore. Since almost all green marks are located in the overlapped zones, we believe that the joint inversions well recover the graphite ore in this area.

Fig. 13 shows the data misfit in the inversions of the survey data. It can be observed that both the individual and joint inversions converge well with the joint inversion running a bit slowly for the reason discussed earlier.

V. Conclusion

In this article, we have successfully developed the joint inversion of AEM and AirMag data based on the LPCC. Compared with the separated inversions, the joint inversion in this article can effectively combine the advantages of the high horizontal resolution of the AirMag method and high vertical resolution and large depth of the AEM method, so that the resolutions of both methods are improved. Analyzing the influence of the subdomain size on the joint inversion results, we found that a too-small subdomain can easily lead to instability in the inversions, while a too-big subdomain size can lead to the reduction of the resolution. Presently, an appropriate subdomain size can only be determined by experiments. We hope to be able to develop an algorithm for adaptive selection of subdomain size, which will be our future research focus.

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